Experiment 7

Ex. No.: 7

Comparison of Two Files for Forensics Investigation by Compare IT Software

Aim

To compare two text files, locate their differences, and confirm whether their contents are identical.

Tools required

1. Compare It on Windows for the manual workflow.
2. macOS command-line equivalents: `diff`, `cmp`, and `shasum`.
3. Two synthetic text files created for this report.

Idea/Theory

1. A side-by-side comparison highlights added, removed, and changed text.
2. A byte comparison identifies the positions at which two files differ.
3. Cryptographic digests provide a compact integrity check: different SHA-256 values prove the byte streams are not identical.
4. File comparison supports verification, but the examiner must preserve the originals and record the tool and options used.

Experiment 7 — continued

Procedure/Steps

1. Install and open Compare It on Windows.
2. Select the first file, then open the second file in the comparison window.
3. Review the coloured difference bars and the highlighted changed lines.
4. Generate the printable difference report if required by the lab.
5. For the executed macOS equivalent, run the commands below from `evidence/exp07/`.

```
diff -u original.txt examined.txt
cmp -l original.txt examined.txt
shasum -a 256 original.txt examined.txt
```

Executed on macOS using diff, cmp, and shasum on sample files created for this report (the manual's Windows tool: Compare It).

Experiment 7 — continued

Observation/Output

```
--- original.txt        2026-10-28 10:40:49
+++ examined.txt        2026-10-28 10:41:53
@@ -1,4 +1,4 @@
 Incident reference: CDF-LAB-SAMPLE
 Evidence item: training-note.txt
 Status: collected
-Finding: No anomaly recorded.
+Finding: No anomaly reported.

   109 143 160
   112 144 164

SHA-256:
35905b3a85d29eee533b83a6dbf37feb02f631083e404eaff9096558a14ec22e  original.txt
0ec1994dd75f09ddcff447d0ab8f746626c6b299baaa827a79d9663a3a126958  examined.txt
```

Result

The real `diff` output identified the changed line, `cmp` reported two differing byte positions, and the SHA-256 digests differed. This demonstrated that the two synthetic sample files were not identical.